

House Price Prediction Using Machine Learning

Project Overview

The objective of this project was to develop a machine learning model capable of predicting house prices based on various property characteristics such as area, number of bedrooms, bathrooms, parking availability, furnishing status, and other housing features. The Housing Prices Dataset from Kaggle was used for the analysis.

Data Preparation

The dataset was loaded using Pandas and examined for missing values, duplicates, and data quality issues. Duplicate records were removed, and categorical variables such as furnishing status, main road access, air conditioning, and preferred area were converted into numerical form using one-hot encoding. This preprocessing step ensured that the machine learning algorithms could effectively utilize all available information.

Model Development

Two regression models were trained and evaluated:

1. Linear Regression
2. Random Forest Regressor

The dataset was divided into training and testing sets using an 80:20 ratio. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Key Findings

The analysis revealed that property area was the most influential factor affecting house prices. Other important features included the number of bathrooms, air conditioning availability, parking spaces, and whether the house was located in a preferred area. These features showed strong relationships with the target variable (price).

Among the two models, the Random Forest Regressor generally achieved better predictive performance than Linear Regression. This indicates that house prices are influenced by complex and non-linear relationships that are captured more effectively by ensemble learning methods.

Visual Analysis

Several visualizations were created to better understand the data:

- A histogram showed the distribution of house prices.
- A correlation heatmap highlighted relationships between features and price.
- An actual-versus-predicted scatter plot demonstrated the accuracy of the model predictions.

Conclusion and Recommendation

The project successfully demonstrated how machine learning can be used to estimate housing prices. Property size, location-related factors, and amenities were found to be the most important drivers of value. Real estate businesses can use such predictive models to support pricing decisions, identify high-value properties, and provide more accurate market estimates to buyers and sellers. Implementing data-driven pricing strategies can improve decision-making and increase customer confidence in property valuations.